KrishiBondhu AI - Initial System Architecture Diagram

Frontend -> Backend API -> Database, with AI and external services | Team CSE4104-7G-T03

Users

- Farmer / Student
- Agricultural Advisor
- Admin

Browser

FRONTEND (Presentation Layer)

React single-page web app, responsive (desktop / tablet / mobile)

- Auth & Profile Pages
- AI Chat Interface (Bangla / English)
- Crop Image Upload & Scan Result
- Fertilizer & Crop Recommendation Forms
- Weather Dashboard
- Pest Library (browse / search)
- Farm Records & Reminders
- Saved Recommendations
- Admin / Advisor Dashboard
- Language Switch (i18n) & Notifications

HTTPS REST/JSON

+ JWT

BACKEND API (Application Layer)

Node.js + Express REST API

Middleware

JWT Authentication | Role-Based Access Control
Input Validation | Rate Limiting | Error Handling

Route Controllers / Modules

- Auth & User
- AI Chat
- Disease Detection
- Fertilizer & Crop Recommendation
- Weather
- Pest & Crop Knowledge Base
- Farm Records
- Reminders
- Admin & Advisor

Services

- AI Service (Gemini client, prompt builder)
- Weather Service (API client + cache)
- Image Upload Handler (validation, resizing)
- Reminder Scheduler (periodic due-check job)
- Logging & Activity Log Service
- Report Generation Service

Stateless server - can be scaled horizontally

API keys & secrets kept in server environment variables

ODM queries

DATABASE (Data Layer)

MongoDB (document database)

- users
- chat_sessions / chat_messages
- feedback
- farm_records / farm_notes
- reminders
- disease_scans
- recommendations / saved_recommendations
- crops / pests / crop_pests
- activity_logs

read / write

IMAGE STORAGE

Uploaded crop images

Server file system / cloud object storage
(image URL saved in disease_scans)

SMTP (password reset)

prompts / images <-> responses

HTTP requests <-> weather data

EXTERNAL SERVICES

AI, weather and email - accessed only through the backend

- **Google Gemini API** — chat answers, image analysis, recommendations
- **Weather API (e.g. OpenWeatherMap)** — current weather & short-term forecast
- **Email Service (SMTP, e.g. Nodemailer)** — password-reset emails

Notes

Deployment platform will be finalised in the development phase. All AI calls go through the backend; the client never sees API keys.